

MTU_ValueService **Technical Documentation**

Electronics
BlueLine with Display 10
and Display 12
for Series 4000 FPP
ADEC (Classic/Comfort)

Functional Description
Operating Instructions
Workshop Manual
Installation and
Commissioning Instructions

System documentation
E532286/01E



Printed in Germany

© 2011 Copyright MTU Friedrichshafen GmbH

This Publication is protected by copyright and may not be used in any way whether in whole or in part without the prior written permission of MTU Friedrichshafen GmbH. This restriction also applies to copyright, distribution, translation, microfilming and storage or processing on electronic systems including data bases and online services.

This handbook is provided for use by maintenance and operating personnel in order to avoid malfunctions or damage during operation.

Subject to alterations and amendments.

Commissioning Note

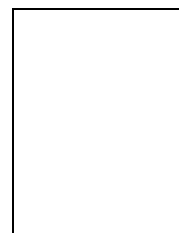
Important

Please complete and return the “Commissioning Note” card below to MTU Friedrichshafen GmbH.

The Commissioning Note information serves as a basis for the contractually agreed logistic support (warranty, spare parts, etc.).



Postcard
MTU Friedrichshafen GmbH
Technical Information Management
Dept. VOT
88040 Friedrichshafen
GERMANY



Commissioning Note

Please use block capitals!



Engine No.:	MTU works order no.:
Engine model:	Date put into operation:
Application : * ☐ Marine ☐ Rail ☐ Genset ☐	Type : Manufacturer:
End user`s address:	
Remarks:	

**Commissioning
Note**

*
Tick appropriate box

1	General Provisions	11
1.1	Safety	11
1.1.1	General conditions	11
1.1.2	Personnel and organizational requirements	12
1.1.3	Safety regulations for startup and operation	13
1.1.4	Safety regulations for maintenance and repair work	14
1.1.5	Conventions for safety instructions in the text	16
2	Functional Description	17
2.1	Product Summary	17
2.1.1	BlueLine system – Overview	17
2.1.2	BlueLine subsystems – Use	19
2.2	Subsystems	20
2.2.1	Remote Control System RCS-5	20
2.2.1.1	RCS-5 BlueLine – Overview	20
2.2.1.2	RCS-5 BlueLine – Use of devices	22
2.2.1.3	RCS-5 BlueLine – Main control stand 1	26
2.2.1.4	RCS-5 BlueLine – Main control stand 2	29
2.2.1.5	RCS-5 BlueLine – Slave control stand	30
2.2.2	Monitoring and Control System MCS-5	32
2.2.2.1	MCS-5 BlueLine – Overview	32
2.2.2.2	MCS-5 BlueLine – Use of devices	34
2.2.2.3	RS 422 serial interface	37
2.2.2.4	MCS-5 BlueLine – Main control stand	41
2.2.2.5	MCS-5 BlueLine – Slave control stand	44
2.2.3	Engine Control System ECS-5	45
2.2.3.1	ECS-5 BlueLine – Overview	45
2.2.3.2	ECS-5 BlueLine – Use of devices	50
3	Operating Instructions	55
3.1	Monitoring and Control System MCS-5	55
3.1.1	Controls and Displays	55
3.1.1.1	MCS-5 – Controls and displays	55
3.1.1.2	LOP display – Overview of screen pages	58
3.1.2	Operation	68
3.1.2.1	Overall BlueLine system – Switching on	68
3.1.2.2	Brightness of indicators/instruments – Adjustment	69
3.1.2.3	Starting the engine	70
3.1.2.4	Override function – Activation	71
3.1.2.5	Override function – Deactivation	72
3.1.2.6	Stopping the engine	73
3.1.2.7	Emergency engine shutdown	74
3.1.2.8	Overall system – Switching off	75
3.1.2.9	PPC oil priming pump control unit – Controls and displays	76
3.1.2.10	Priming Pump Controller PPC – Function check	77
3.2	Remote Control System RCS-5	78
3.2.1	Controls and Displays (Standard Version)	78
3.2.1.1	Command unit for one shaft	78

3.2.1.2	Command unit for two shafts.....	80
3.2.1.3	Command unit for three shafts.....	82
3.2.1.4	Command unit for four shafts.....	84
3.2.2	Controls and Displays (Special Version).....	86
3.2.2.1	Operating station and rotary encoder modules	86
3.2.3	Operation.....	88
3.2.3.1	Initial command – Allocation.....	88
3.2.3.2	Command transfer.....	89
3.2.3.3	Operation.....	90
3.2.3.4	Changing engine speed without engaging.....	91
3.2.3.5	Single Control Lever mode (SCL) – Activation.....	92
3.2.3.6	Single Control Lever mode (SCL) – Deactivation.....	93
3.2.3.7	Trolling mode – Activation.....	96
3.2.3.8	Trolling mode – Deactivation.....	97
3.3	Engine Control System ECS-5.....	98
3.3.1	Controls and Displays	98
3.3.1.1	Local Operating Panel LOP.....	98
3.3.1.2	Local Operating Station LOS and Local Operating Panel LOP – Controls and displays.....	99
3.3.2	Normal Operation: LOS or LOP with Display.....	101
3.3.2.1	Engine – Switching ready for operation.....	101
3.3.2.2	Local mode – Activation.....	102
3.3.2.3	Starting the engine	103
3.3.2.4	Local mode – Gearbox engagement/disengagement.....	104
3.3.2.5	Local mode – Engine speed adjustment.....	105
3.3.2.6	Engine – Shutdown	106
3.3.2.7	Local mode – Deactivation	107
3.3.2.8	Engine start – Interlock	108
3.3.2.9	Engine – Emergency shutdown.....	109
3.3.3	Operation during Servicing: LOP without Display	110
3.3.3.1	Starting the engine	110
3.3.3.2	Stopping the engine	111
3.3.3.3	Engine start – Interlock	112
3.3.3.4	Engine – Emergency shutdown.....	113
3.4	Troubleshooting.....	114
3.4.1	Faults in BlueLine system.....	114
3.4.2	Troubleshooting.....	115
3.4.3	Alarms – General information.....	119
3.4.4	Single-point alarms – Acknowledgment.....	120
3.4.5	Fault messages on the LOP display.....	121
3.4.6	Fault indication on printed circuit board MPU 29.....	127
3.4.7	Fault messages – Engine Control Unit.....	131
3.4.8	Oil priming pump PPC – Troubleshooting.....	181
3.5	Task Description.....	182
3.5.1	Wiring (General) for Engine/Gearbox/Unit.....	182
3.5.1.1	Engine wiring – Check	182
3.5.2	Control and Display Elements, General.....	183
3.5.2.1	Indicator lamp – LED replacement.....	183
4	Workshop Manual.....	185

4.1	Task Description.....	185
4.1.1	Operating Tests	185
4.1.1.1	Test mode, BlueLine – Execution	185
4.1.1.2	Overspeed test	188
4.1.1.3	Emergency stop test	189
4.1.1.4	Cranking	190
4.1.2	Checks and Settings	191
4.1.2.1	Serial interface terminator plugs – Check	191
4.1.2.2	Jumper configuration of MCS-5 control unit – Check	193
4.1.2.3	CAN bus line resistance – Check	194
4.1.2.4	LOP power supply – Check	196
4.1.2.5	Jumper configuration of LOP – Check	198
4.1.2.6	Control stand configuration in Local Operating Panel LOP – Check	200
4.1.2.7	Shaft number setting – Check	202
4.1.2.8	LOP-DIS settings – Check	203
4.1.2.9	LOP display DIS – Setting	204
4.1.2.10	PIM 4 node number – Check	205
4.1.2.11	CAN bus connections PIM 4 – Check	206
4.1.2.12	GCU 3 jumper configuration – Check	208
4.1.2.13	GCU 6 jumper configuration – Check	209
4.1.2.14	GCU node number – Check	210
4.1.2.15	MPU 29 node address – Check	211
4.1.2.16	MPU 29 node address – Setting	212
4.1.2.17	MPU 29 node address in LOP – Setting	213
4.1.2.18	MPU 29 node address in LOP – Check	214
4.1.2.19	Configuration	215
4.1.3	Repair Work	217
4.1.3.1	PIM 4 cover – Removal	217
4.1.3.2	PIM 4 cover – Installation	218
4.1.3.3	Command unit – Replacement	219
4.1.3.4	Rotary encoder module – Replacement	220
4.1.3.5	Operating station – Replacement	221
4.1.3.6	Analog display instrument VDO OceanLine – Replacement	222
4.1.3.7	Lamp in analog display instrument VDO OceanLine – Replacement	223
4.1.3.8	Indicator lamp – Replacement	224
4.1.3.9	Indicator lamp – Removal	225
4.1.3.10	Indicator lamp – Installation	226
4.1.3.11	Indicator lamp – LED replacement	227
4.1.3.12	Contactorm – Replacement	229
4.1.3.13	LED in pushbutton/switch – Replacement	230
4.1.3.14	PIM 4 parameters – Reset	232
4.1.3.15	Fuse in control unit PIM 4 – Replacement	233
4.1.3.16	Control unit PIM 4 – Replacement	234
4.1.3.17	Printed circuit board in PIM cassette – Replacement	236
4.1.3.18	Fuse on printed circuit board in PIM cassette – Replacement	238
4.1.3.19	Printed circuit board MPU 29 – Removal	241
4.1.3.20	Printed circuit board MPU 29 – Installation	242
4.1.3.21	Printed circuit board MPU 29 – Replacement	243
4.1.3.22	Data module MEM on MPU 29 – Replacement	244
4.1.3.23	Data module MEM on MPU 29 – Transfer	245

4.1.3.24	LOP fuse – Replacement.....	246
4.1.3.25	LOP with display front flap – Replacement.....	248
4.1.3.26	LOP – Replacement.....	250
4.1.3.27	Local Operating Station LOS – Replacement.....	251
4.1.3.28	Gear Control Unit GCU – Replacement.....	252
4.1.3.29	Priming Pump Controller PPC – Repair.....	254
4.1.3.30	Priming Pump Controller PPC – Replacement.....	256
4.1.4	Supplementary Technical Information.....	257
4.1.4.1	PIM 4 RCS – Internal design.....	257
4.1.4.2	Control unit PIM 4 MCS – Internal design.....	262
4.1.4.3	Local Operating Panel LOP – Design.....	268
4.1.4.4	Gear Control Unit GCU – Design.....	277
4.1.4.5	Priming Pump Controller PPC – Design.....	282
5	Installation and Commissioning Instructions.....	283
5.1	Installation.....	283
5.1.1	Preparation.....	283
5.1.1.1	Cable routes and openings between all installation locations – Check ..	283
5.1.1.2	Cables – Routing between installation locations.....	284
5.1.2	Mechanical Installation.....	286
5.1.2.1	PIM 4 module housing – Installation.....	286
5.1.2.2	OceanLine analog display instruments – Installation.....	287
5.1.2.3	Horn VDO – Installation.....	289
5.1.2.4	Indicator lamp – Installation.....	290
5.1.2.5	Illuminated pushbuttons/key switches – Installation.....	291
5.1.2.6	GCU – Installation.....	293
5.1.2.7	Priming Pump Controller PPC – Installation.....	295
5.1.2.8	LOP – Installation.....	296
5.1.2.9	LOS – Installation.....	298
5.1.2.10	Command unit ROS 11 – Installation.....	300
5.1.2.11	Command unit ROS 7 / ROS 9 – Installation.....	301
5.1.2.12	Rotary encoder modules ROS 10/P and ROS 10/S – Installation.....	303
5.1.2.13	Remote Operating Station ROS 10/T – Installation.....	304
5.1.2.14	Terminal box X010 – Installation.....	306
5.1.3	Electrical Installation.....	308
5.1.3.1	HSK cable gland – Preparation.....	308
5.1.3.2	Terminal block – Preparation.....	309
5.1.3.3	PIM 4 cover – Removal.....	310
5.1.3.4	GCU 3 – Connection.....	311
5.1.3.5	GCU – Pin assignment for non-MTU gearbox.....	313
5.1.3.6	Engine Control Unit ECU 7 – Connection.....	315
5.1.3.7	Display instruments on main control stand – Connection.....	316
5.1.3.8	Controls and status indicator lamps on main control stands – Connection.....	317
5.1.3.9	Indicator lamps for single-point alarms – Connection.....	318
5.1.3.10	Devices on a slave control stand – Connection.....	319
5.1.3.11	PIM 4 MCS control unit – Connection.....	320
5.1.3.12	PPC – Connection (single phase).....	323
5.1.3.13	PPC – Connection (three phases).....	324
5.1.3.14	Starter and battery-charging generator – Connection.....	325
5.1.3.15	Barring tool limit switch – Connection.....	329

	5.1.3.16	LOP display (option), main control stand – Connection.....	330
	5.1.3.17	LOP – Connection	331
	5.1.3.18	Additional Yard signals (optional) – Connection.....	334
	5.1.3.19	LOS – Connection	336
	5.1.3.20	Command units on triple-shaft propulsion systems – Connection.....	337
	5.1.3.21	Command units on propulsion systems with four shafts – Connection...	339
	5.1.3.22	Command unit ROS 7 / ROS 9 – Connection.....	340
	5.1.3.23	Rotary encoder module and remote operating station – Connection.....	341
	5.1.3.24	PIM 4 RCS control unit – Connection.....	342
	5.1.3.25	RCS extension module – Installation in PIM 4 RCS	344
	5.1.3.26	Terminal box X010 – Connection (battery-charging generator charges starter battery)	345
	5.1.4	Data Transmission	349
	5.1.4.1	RS422 – Transmission values.....	349
	5.1.5	Technical Data.....	350
	5.1.5.1	ECU 7 – Technical data	350
	5.1.5.2	GCU 3 – Technical data.....	352
	5.1.5.3	GCU 6 – Technical data.....	353
	5.1.5.4	PPC – Technical data.....	354
	5.1.5.5	LOP 8 – Technical data.....	355
	5.1.5.6	Technical data X010	356
	5.1.5.7	PIM 4 control unit – Technical data.....	357
	5.1.5.8	Display DIS 10 – Technical data.....	358
	5.1.5.9	Display DIS 12 – Technical data.....	360
5.2		Initial Operation.....	362
	5.2.1	Preparation.....	362
	5.2.1.1	Checks prior to startup.....	362
	5.2.1.2	Data module MEM on MPU 29 – Installation.....	363
	5.2.1.3	Operating voltage – Initial connection	364
	5.2.1.4	Supply voltage distribution – Check.....	367
	5.2.2	Settings.....	370
	5.2.2.1	Configuration.....	370
	5.2.2.2	Settings via Mini Dialog.....	372
	5.2.3	Tests.....	387
	5.2.3.1	Initial engine start-up at the service interface of the LOP.....	387
	5.2.3.2	Initial engine start-up at the display on a Local Operating Panel LOP or a Local Operating station LOS.....	389
	5.2.3.3	Initial engine start at MCS.....	391
	5.2.3.4	Speed control and clutch setting – Check.....	393
	5.2.3.5	Command transfer – Check (only when several control stands are installed).....	394
	5.2.3.6	System reset.....	395
6		Appendix A.....	397
	6.1	Abbreviations	397
	6.2	Conversion tables.....	400
	6.3	MTU Contact/Service partners.....	404
7		Appendix B.....	405

7.1 Index..... 405